

From shuffling cards to shuffling genes

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Adjacent Transposition

- ▶ Choose an index $i \in \{1, \dots, n-1\}$, and swap the neighbors in positions i and $i+1$.
- ▶ Possible adjacent transpositions $(1\ 2), (2\ 3), \dots, (n-1\ n)$.
- ▶ **Lazy version:** at each step we stay put with probability $1/2$; otherwise (with probability $1/2$) we choose $i \in \{1, \dots, n-1\}$ uniformly at random and perform the swap $(i\ i+1)$

Adjacent Transposition

Definition (Lazy version)

At each step:

1. With probability $1/2$, stay at the current permutation.
2. With probability $1/2$, choose $i \in \{1, \dots, n-1\}$ uniformly and apply the swap $(i \ i+1)$.

Transition probabilities. For permutations $\sigma, \tau \in \mathbb{S}_n$,

$$P(\sigma, \tau) = \begin{cases} \frac{1}{2} & \text{if } \tau = \sigma, \\ \frac{1}{2(n-1)} & \text{if } \tau = \sigma \circ (i \ i+1) \text{ for some } i, \\ 0 & \text{otherwise.} \end{cases}$$

Aperiodic, easier to analyze coupling/mixing.

Coupling Rule

- ▶ We run two copies of the Markov chain in parallel:
 - ▶ a **left** deck σ_t and a **right** deck σ'_t .
- ▶ At each step:
 1. Choose an adjacent edge $(i, i + 1)$ uniformly from $\{(1, 2), \dots, (n - 1, n)\}$.
 2. Flip a fair coin to decide whether the left deck performs the swap.
 3. Look at the two cards currently at locations i and $i + 1$ in both decks, then:
 - ▶ If either $\sigma_t(i) = \sigma'_t(i + 1)$ or $\sigma_t(i + 1) = \sigma'_t(i)$, do the *opposite* on the right deck (transpose if the left stayed still, and vice versa).
 - ▶ Otherwise, perform the *same* action on the right deck as on the left deck.

This forces the two copies of a card to move *toward* each other whenever we choose the edge between them.

Upper bound on mixing time via Coupling

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Definitions

- For a fixed card a , define the **synchronization time**

$$\tau_a := \min\{ t \geq 0 : \text{the positions of card } a \text{ in the two decks are equal at time } t \}.$$

- Let X_t be the **(unsigned) distance** between the positions of card a in the two decks at time t .

The coupling ensures:

$$|X_{t+1} - X_t| \leq 1 \quad \text{and} \quad (t \geq \tau_a) \Rightarrow X_t = 0.$$

Upper bound on Mixing time via Coupling

Let M be the transition matrix of a random walk on the path with vertices $\{0, 1, \dots, n-1\}$:

- ▶ At interior vertices $i \in \{1, \dots, n-2\}$, the walk moves $i \rightarrow i-1$ with probability $1/(n-1)$, and $i \rightarrow i+1$ with probability $1/(n-1)$.
- ▶ From $n-1$, it moves down to $n-2$ with probability $1/(n-1)$.
- ▶ Under all other circumstances, it stays put. In particular, state 0 is absorbing.

For $1 \leq i \leq n-1$,

$$\Pr\{X_{t+1} = i-1 \mid X_t = i, \sigma_t, \sigma'_t\} = M(i, i-1),$$

and since one or both cards can be at an endpoint (which can block increases),

$$\Pr\{X_{t+1} = i+1 \mid X_t = i, \sigma_t, \sigma'_t\} \leq M(i, i+1).$$

Upper bound on mixing time via Coupling

Define a delayed random walk (Y_t) with transition matrix M on $\{0, 1, \dots, n-1\}$:

- ▶ With probability $1 - \frac{2}{n-1}$, do nothing;
- ▶ With probability $\frac{1}{n-1}$, move $+1$ (unless at $n-1$, where it holds);
- ▶ With probability $\frac{1}{n-1}$, move -1 ;
- ▶ Absorb at 0.

We can couple so that $X_0 = Y_0$ and $X_t \leq Y_t$ for all $t \geq 0$.

Let $\tau_a = \min\{t : X_t = 0\}$ be the *synchronization* time for card a , and let $\tau_0^Y = \min\{t : Y_t = 0\}$ be the hitting time of 0 for Y_t . Then

$$\tau_a \leq \tau_0^Y \quad \text{and} \quad \mathbb{E}[\tau_a] \leq \mathbb{E}[\tau_0^Y].$$

Upper bound on Mixing time via Coupling

Double the line. Construct a simple random walk (Z_k) on the doubled line

$$\{0, 1, \dots, 2n - 1\}, \quad Z_0 = n,$$

with *absorption* at $\{0, 2n - 1\}$, and steps ± 1 with probability $1/2$.

Define

$$B_k = \min\{Z_k, 2n - 1 - Z_k\} \quad \text{until absorption.}$$

B_k is absorbing at 0 and *reflecting* at n .

Consequently, the hitting time of 0 for B_k is exactly the hitting time of the boundary set $\{0, 2n - 1\}$ for Z_k :

$$\tau_0^B = \inf\{k : B_k = 0\} = \inf\{k : Z_k \in \{0, 2n - 1\}\} = \tau_{\{0, 2n - 1\}}^Z.$$

Upper bound on Mixing time via Coupling

Gambler's ruin facts. Let $N := 2n - 1$ and $i \in \{0, \dots, N\}$.

$$\mathbb{E}_i \left[\tau_{\{0, N\}}^Z \right] = i(N - i).$$

Starting at n , $\mathbb{E}[\tau_0^B] = \mathbb{E}_n \left[\tau_{\{0, 2n+1\}}^Z \right] = n(n - 1).$

Upper bound on Mixing time via Coupling

Let $K := \tau_0^B$ be the number of moves until absorption.

In the **delayed process**, time is measured in *rounds*. In each round, a move occurs with probability

$$p = \frac{1}{n-1} + \frac{1}{n-1} = \frac{2}{n-1}.$$

Let $G_1, G_2, \dots$ be i.i.d. $\text{Geom}(p)$ random variables (rounds until a move). If K moves are needed to reach absorption, the total rounds is

$$\tau_0^Y = \tau_{\text{rounds}} = G_1 + G_2 + \dots + G_K.$$

Condition on K :

$$\mathbb{E}[\tau_0^Y] = \mathbb{E}[\mathbb{E}[\tau_{\text{rounds}} \mid K]] = \frac{\mathbb{E}[K]}{p}.$$

Upper bound on Mixing time via Coupling

Using $\mathbb{E}[K] \leq n(n-1)$ and $p = \frac{2}{n-1}$:

$$\mathbb{E}[\tau_0^Y] \leq \frac{n(n-1)}{2/(n-1)} = \frac{n(n-1)^2}{2} \leq \frac{n^2(n-1)}{2} < \frac{n^3}{2}$$

$$\mathbb{E}[\tau_a] \leq \mathbb{E}[\tau_0^Y] \Rightarrow \mathbb{E}[\tau_a] < \frac{n^3}{2}$$

By Markov's inequality,

$$\mathbb{P}\{\tau_a > n^3\} < \frac{1}{2}.$$

Upper bound on Mixing time via Coupling

From the block argument, for each card a ,

$$\mathbb{P}(\tau_a > k n^3) \leq 2^{-k}.$$

Choose $k = \lceil 2 \log_2 n \rceil$. Then

$$\mathbb{P}(\tau_a > 2n^3 \lceil \log_2 n \rceil) \leq 2^{-\lceil 2 \log_2 n \rceil} \leq \frac{1}{n^2}.$$

Coupling time. The time until *all* cards are synchronized is

$$\tau_{\text{couple}} := \min\{t \geq 0 : X_t^{(a)} = 0 \text{ for all } a\} = \max_{1 \leq a \leq n} \tau_a.$$

Union bound over cards.

$$\Pr(\tau_{\text{couple}} > 2n^3 \lceil \log_2 n \rceil) = \Pr\left(\max_a \tau_a > 2n^3 \lceil \log_2 n \rceil\right) \leq \sum_{a=1}^n \Pr(\tau_a > 2n^3 \lceil \log_2 n \rceil) < \frac{1}{n}.$$

Upper bound on Mixing time via Coupling

For a coupling (X_t, Y_t) with $X_0 = x$, $Y_0 = y$, let

$$\tau_{\text{couple}} := \min\{t : X_s = Y_s \text{ for all } s \geq t\}.$$

Then, by **Theorem 5.4**,

$$\|P^t(x, \cdot) - P^t(y, \cdot)\|_{\text{TV}} \leq \Pr_{x,y}\{X_t \neq Y_t\} = \Pr_{x,y}\{\tau_{\text{couple}} > t\}.$$

Defining the pairwise distance

$$\bar{d}(t) := \max_{x,y} \|P^t(x, \cdot) - P^t(y, \cdot)\|_{\text{TV}},$$

we obtain

$$\bar{d}(t) \leq \max_{x,y} \Pr_{x,y}\{\tau_{\text{couple}} > t\}.$$

Upper bound on Mixing time via Coupling

$$d(t) := \max_x \|P^t(x, \cdot) - \pi\|_{\text{TV}}, \quad d(t) \leq \bar{d}(t)$$

Consequently,

$$d(t) \leq \max_{x,y} \Pr\{\tau_{\text{couple}} > t\}$$

We get,

$$d(2n^3 \lceil \log_2 n \rceil) < \frac{1}{n}.$$

Mixing time. $t_{\text{mix}}(\varepsilon) := \inf\{t : d(t) \leq \varepsilon\}$ and taking $\varepsilon = \frac{1}{n}$, we conclude

$$\boxed{t_{\text{mix}}\left(\frac{1}{n}\right) \leq 2n^3 \lceil \log_2 n \rceil}.$$

Shuffling Genes

Definition. For $1 \leq i \leq j \leq n$, the reversal $\rho_{i,j} \in S_n$ is the permutation that *reverses* the block of positions $i, i+1, \dots, j$ and leaves everything else fixed. Formally,

$$\rho_{i,j}(k) = \begin{cases} k, & k < i \text{ or } k > j, \\ i+j-k, & i \leq k \leq j. \end{cases}$$

(When $i = j$, $\rho_{i,i}$ is the identity.)

Example. Starting from $[1, 2, 3, 4, 5, 6, 7]$, applying $\rho_{3,6}$ reverses positions 3–6:

$$[1, 2, \boxed{3, 4, 5, 6}, 7] \xrightarrow{\rho_{3,6}} [1, 2, \boxed{6, 5, 4, 3}, 7].$$

The L -reversal chain on S_n

Circular set-up. To avoid edge effects we view positions on a cycle (indices taken modulo n). A segment of length k starting at i is $\{i, i+1, \dots, i+k\}$

Definition (L -reversal chain). Fix a function $L = L(n)$ with $1 \leq L(n) \leq n$. From the current permutation $\sigma \in S_n$, perform one step as follows:

1. Choose a start $i \in \{1, \dots, n\}$ uniformly.
2. Choose a length $k \in \{1, \dots, L\}$ uniformly.
3. Apply the reversal $\rho_{i,i+k}$ to σ (indices mod n).

There are nL equally likely moves (reversals of circular segments of length $\leq L$); the choice $k = 1$ is the identity move, so

$$\Pr\{\text{id move}\} = \frac{n}{nL} = \frac{1}{L} \Rightarrow \text{the chain is lazy/aperiodic.}$$

Basic properties.

- ▶ *State space*: all $n!$ permutations S_n .
- ▶ *Reversibility*: each reversal is its own inverse ($\rho_{i,j} = \rho_{i,j}^{-1}$) $\Rightarrow$ transition matrix is symmetric, so the uniform distribution on S_n is stationary and the chain is reversible.

Lower Bound on Mixing Time

Proposition 16.2

Consider the family of L -reversal chains with $1 \leq L(n) < n/2$. For every $\varepsilon \in (0, 1)$, there exists c_ε such that

$$t_{\text{mix}}(\varepsilon) \geq \frac{n \log n}{2} - c_\varepsilon n, \quad \text{as } n \rightarrow \infty.$$

Lower Bound on Mixing Time for the L -Reversal Chain

A single reversal can cut at most two edges, so we track the number of preserved adjacencies.

$$\sigma(k+1) - \sigma(k) = \pm 1 \pmod{n} \Rightarrow \text{edge } (k, k+1) \text{ is } \mathbf{conserved}.$$

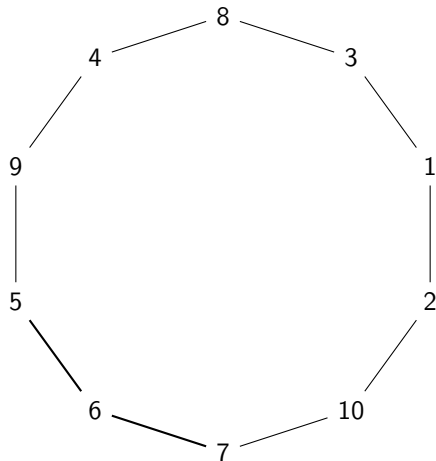
Under the uniform measure π on S_n ,

$$\Pr_{\pi}\{\text{edge } (k, k+1) \text{ is conserved}\} = \frac{2}{n-1}.$$

Hence, the expected number of conserved edges under π is

$$\mathbb{E}_{\pi}[\# \text{ conserved edges}] = n \cdot \frac{2}{n-1} = 2 + o(1), \quad \text{as } n \rightarrow \infty.$$

Conserved edges for a permutation on a circle



The permutation 1, 3, 8, 4, 9, 5, 6, 7, 10, 2 has three conserved edges (bold): (1,2), (5,6), and (6,7).

Lower Bound on Mixing Time for the L -Reversal Chain

A cycle edge $(k, k + 1)$ is called **undisturbed** by time t if it has not been cut by any reversal. Let $U(t)$ be the number of *undisturbed* edges after t steps:

$$U(t) = \#\{\text{cycle edges not cut by any reversal up to time } t\} = \sum_{k=1}^n U_k(t).$$

$U_k(t)$ is the indicator that edge $(k, k + 1)$ is undisturbed at time t .

A fixed edge is cut in each step with probability $\frac{2}{n}$, so

$$\mathbb{E}[U(t)] = n \left(1 - \frac{2}{n}\right)^t \approx ne^{-2t/n}.$$

Choosing $t = \frac{n}{2} \log n - c_\epsilon n$,

$$\mathbb{E}[U(t)] \rightarrow e^{2c_\epsilon} \quad \text{as } n \rightarrow \infty.$$

Variance of $U(t)$ via indicators

A fixed edge is hit by exactly $2L$ reversals. Success probability,

$$p := \Pr(U_i = 1) = \left(1 - \frac{2}{n}\right)^t.$$

Joint survival for $i \neq j$ (cyclic distance):

$$\Pr(U_i = 1, U_j = 1) = \begin{cases} \left(\frac{nL - (4L - 1)}{nL}\right)^t, & 1 \leq \text{dist}(e_i, e_j) \leq L, \\ \left(\frac{nL - 4L}{nL}\right)^t, & \text{dist}(e_i, e_j) > L. \end{cases}$$

If $\text{dist}(e_i, e_j) > L$ then

$$\left(1 - \frac{4}{n}\right)^t \leq \left(1 - \frac{2}{n}\right)^{2t} = p^2 \Rightarrow \text{Cov}(U_i, U_j) \leq 0.$$

Bounding $\text{Var}(U(t))$

Variance decomposition:

$$\text{Var}(U(t)) = \sum_{i=1}^n \text{Var}(U_i) + \sum_{i \neq j} \text{Cov}(U_i, U_j), \quad \text{Var}(U_i) = p(1-p).$$

Only *near* pairs ($\text{dist} \leq L$) can have positive covariance;

$$\text{Var}(U(t)) \leq np(1-p) + 2nL \left(\left(1 - \frac{4 - \frac{1}{L}}{n}\right)^t - p^2 \right).$$

Mean value theorem bound. With $t = \frac{n}{2} \log n$,

$$2nL \cdot \frac{t}{nL} \left(1 - \frac{3}{n}\right)^{t-1} = n \log n \cdot O(n^{-3/2}) = o(1).$$

Hence,

$$\text{Var}(U(t)) \leq [1 + o(1)] \mathbb{E}[U(t)].$$

From variance control to a TV lower bound

Let

$$A = \left\{ \sigma \in S_n : \# \text{conserved edges in } \sigma \geq \frac{1}{2} \mathbb{E}[U(t)] \right\}.$$

Under the uniform measure π on S_n , Markov's inequality gives

$$\pi(A) \leq \frac{5}{\mathbb{E}[U(t)]}.$$

By Chebyshev,

$$P^t(\text{id}, A^c) = \Pr\left(|U(t) - \mathbb{E}[U(t)]| > \frac{1}{2} \mathbb{E}[U(t)]\right) \leq \frac{\text{Var}(U(t))}{(\mathbb{E}[U(t)]/2)^2} < \frac{5}{\mathbb{E}[U(t)]}.$$

Total variation lower bound

By the definition of total variation distance,

$$\|P^t(\text{id}, \cdot) - \pi\|_{\text{TV}} \geq P^t(\text{id}, A) - \pi(A) \geq \left(1 - \frac{5}{\mathbb{E}[U(t)]}\right) - \frac{5}{\mathbb{E}[U(t)]} = 1 - \frac{10}{\mathbb{E}[U(t)]}.$$

$$d(t) \geq \|P^t(\text{id}, \cdot) - \pi\|_{\text{TV}} \geq 1 - \frac{10}{\mathbb{E}[U(t)]}.$$

At the chosen time $t = \frac{n}{2} \log n - c_\varepsilon n$ we have

$$\mathbb{E}[U(t)] \rightarrow e^{2c_\varepsilon},$$

so for large enough c_ε ,

$$d(t) > \varepsilon.$$

Therefore,

$$t_{\text{mix}}(\varepsilon) \geq \frac{n}{2} \log n - c_\varepsilon n.$$